

UP Maternal Monitoring Platform

Product Brief



Purpose	A statewide maternal care coordination, risk monitoring, and accountability platform for pregnancy tracking, HRP escalation, delivery preparedness, and maternal death intelligence.

1. Executive Summary

The UP Maternal Monitoring Platform is proposed as a statewide digital continuum-of-care system to identify every pregnancy early, route the case to the correct frontline public health network, maintain a single longitudinal maternal record, monitor routine and high-risk pregnancies, ensure timely ANC follow-up, enable referral readiness, and support reduction in preventable maternal deaths. This initiative is aligned with the **Sustainable Development Goals (SDGs)** by strengthening early pregnancy tracking, timely risk detection, and coordinated maternal care, thereby contributing to the saving of both **mothers' and infants' lives**.

The platform is intended to function both as an operational care coordination system and as a state-level monitoring and accountability instrument. It is not a generic hospital app; it is a maternal surveillance, tracking, escalation, and preparedness platform designed for Uttar Pradesh.

2. Strategic Objective

- Capture pregnancy at the earliest possible stage through citizen, family, ASHA, ANM, or facility-assisted registration.
- Auto-route each case to the mapped ASHA, ANM, and facility chain based on geography.
- Maintain one single maternal case record across the pregnancy journey, without duplicate identifiers.
- Store ANC records date-wise in one timeline, with due, missed, and rescheduled workflows.
- Flag High-Risk Pregnancy (HRP) cases early and place them on an enhanced monitoring pathway.
- Trigger delivery preparedness, referral, and transport readiness in advance of expected deliveries.
- Measure maternal deaths, near-miss events, referral delays, and missed follow-up points for leadership review.

3. Operating Model

The platform begins with a minimal registration layer. A beneficiary, husband, family member, ASHA, ANM, facility staff, or helpline operator can submit basic details such as name, mobile number, village, block, and district. Once submitted, the system automatically identifies the mapped ASHA and ANM of that area and assigns the case to them for verification and detailed case creation.

Thereafter, ASHA, ANM, and facility staff progressively enrich the same case record with pregnancy details, clinical updates, ANC history, risk flags, referrals, transport needs, and delivery outcomes. Each new update is reflected across the care chain according to role-based visibility rules.

4. Identifier Strategy

The platform should avoid creating multiple IDs for the same pregnancy. The recommended hierarchy is: existing RCH or beneficiary tracking ID where available; an ABHA-linked profile where available; and only then a single system-generated fallback reference that it can be brainstormed on. Any fallback reference must remain the same throughout the life of that pregnancy and must never be duplicated across visits, referrals, or facilities.

5. End-to-End Workflow

Stage	What Happens	Responsible Actor(s)
Pregnancy identification	Basic details entered through citizen, family, worker, or facility interface.	Beneficiary / Family / ASHA / ANM / Facility

Auto-assignment	Case routed to mapped ASHA, ANM, and facility chain.	System
Verification	Detailed pregnancy profile verified and completed.	ASHA / ANM
ANC timeline	ANC 1 to ANC 9 and additional contacts are recorded date-wise.	ANM / Facility
Risk flagging	HRP logic runs after every relevant update.	System + Clinician
Delivery preparedness	30-day pre-delivery readiness workflow begins for HRP cases.	System + All stakeholders
Outcome capture	Delivery, near-miss, maternal death, and postpartum status recorded.	Facility / District

6. HRP Monitoring and ANC Logic

A key strength of the platform is active high-risk pregnancy monitoring. HRP may be flagged based on severe anaemia, hypertension, previous C-section, diabetes, multiple pregnancy, previous obstetric complication, repeated missed ANC visits, clinical judgment, or other programme-defined criteria.

Once flagged, HRP cases should automatically move into an enhanced monitoring pathway. In this pathway, the platform should support a minimum of 9 ANC milestones in the monitoring workflow, increased visibility across the care chain, and stronger facility and transport preparedness before delivery.

- Each ANC entry should create a date-stamped update in the same timeline.
- The system should auto-calculate the next due ANC date.
- Before an ANC due date, reminders should be sent to the beneficiary and all relevant health workers.
- If an ANC is missed, the case should move to Missed ANC status, the reason should be recorded, and a revised date should be assigned.
- Nudges should continue until the follow-up is completed or formally closed.

7. Mandatory 30-Day HRP Pre-Delivery Alert Rule

For every High-Risk Pregnancy (HRP) case, the system shall trigger a mandatory pre-delivery alert exactly 30 days before the Expected Delivery Date (EDD). This alert must inform all relevant stakeholders, including transport or ambulance services, and require acknowledgement, readiness marking, or escalation where needed.

The 30-day HRP alert must be sent to the following stakeholders:

- Pregnant woman / beneficiary
- Husband or family contact
- ASHA
- ANM
- Mapped PHC
- Mapped CHC or higher referral facility
- Designated doctor / obstetric team

- Ambulance or transport desk, including 102/108 integration where applicable
- District nodal monitoring team for critical HRP cases

The alert payload should contain beneficiary identity, risk category, HRP reason, EDD, mapped delivery facility, referral destination if needed, transport readiness requirement, and the next required action. Critical stakeholders should be required to acknowledge receipt; if acknowledgement or readiness is not received within the defined time, the case should escalate to the next supervisory level.

8. Accountability Framework

The platform should make every important action traceable. A due ANC, a missed ANC, an HRP alert, a referral acknowledgement, a transport readiness action, and a delivery preparedness action should each have a visible owner, due date, completion date, and escalation status.

Action Point	Visible Owner	Status Needed	Escalation Trigger
ANC due follow-up	ASHA / ANM	Completed / Missed / Rescheduled	Not completed by due date
HRP acknowledgement	ANM / Facility / Doctor	Acknowledged / Pending	No acknowledgement within SLA
Referral acceptance	Receiving facility	Accepted / Pending	No facility acknowledgement
Transport readiness	Ambulance/transport desk	Ready / Issue raised	No readiness confirmation
30-day delivery preparedness	Facility + field chain	Ready / Pending	No closure before delivery window

9. Maternal Death and Near-Miss Measurement

The platform should support both prospective maternal risk surveillance and serious outcome measurement. It should capture suspected and confirmed maternal deaths, maternal near-miss events, location and date of event, ANC history before outcome, referral path, transport history, and likely delay points or missed opportunities.

- Total registered pregnancies
- Total HRP pregnancies
- ANC completion levels
- Missed ANC burden and reschedule compliance
- District-wise and facility-wise risk concentration
- Maternal death count among registered beneficiaries
- Maternal near-miss count
- Referral delay patterns and transport bottlenecks
- Facility clustering of serious adverse outcomes

10. Functional Modules

Beneficiary / Family: Simple registration, case tracking, reminders, delivery guidance, emergency contact actions.

ASHA / ANM: Case verification, home visits, ANC updates, counselling, referral initiation, missed ANC follow-up.

Facility: Registry, referral inbox, HRP queue, doctor alert center, delivery readiness board, outcome capture.

Transport: Alert queue, readiness confirmation, trip assignment and tracking.

District / State: Operational dashboards, escalation queues, maternal death and near-miss analytics, hotspot views.

11. Design Directions

The design team should create a responsive experience for mobile, tablet, and desktop. Mobile should focus on beneficiary and worker workflows with large touch targets, low-literacy support, voice input, and prominent call actions. Desktop and tablet should emphasize registries, dashboards, maps, referral queues, and accountability panels.

The solution architecture team should design the platform as a modular, scalable, cloud-based application with API-driven services, role-based access control, configurable business rules for ANC and HRP logic, event-based notifications, audit trails, and future integration readiness with existing public health systems.

12. Recommended Deliverables

- Leadership presentation summarizing objectives, workflow, value, and implementation roadmap.
- Responsive UI and wireframe pack for citizen, worker, facility, district, and state users.
- Solution architecture diagram, data model, and workflow logic document.
- Role-access matrix and accountability matrix.
- Phased implementation plan with pilot and statewide scale-up approach.

13. Closing Note

The UP Maternal Monitoring Platform should be positioned as a statewide maternal risk governance platform. Its core value lies in making every pregnancy visible, every HRP case actionable, every missed follow-up traceable, every 30-day pre-delivery readiness action accountable, and every serious maternal outcome measurable.